



YAKWETU™

Climate-Based Variable Pricing Project

Predicting Content Demand & Dynamic Pricing
with Climate-Driven Analytics

Group Members

Shallom Githui

Florence Joy Aran

Simon Musyoki

Steven Joel



Project overview

This project explores how weather influences movie viewing behavior on YAKWETU™'s MyMovies.Africa™ platform. By combining climate data with user activity and transaction records, we predict high-demand periods and implement a dynamic pricing model. The goal is to personalize user experiences, boost revenue, and align content pricing with forecasted demand.



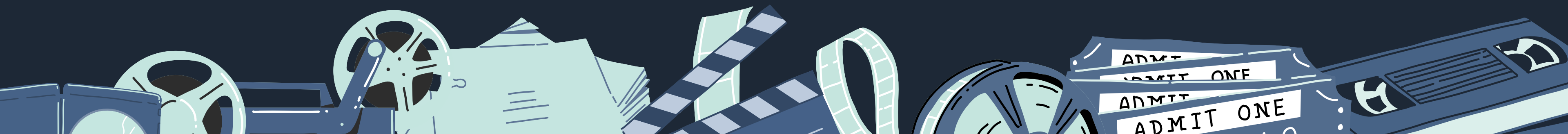
Problem statement

As MyMovies.Africa™ expands, YAKWETU™ Online Ltd. seeks to make smarter, data-driven pricing decisions. Since weather can impact user mood and viewing habits, this project tackles the challenge of developing a flexible pricing model that adjusts to climate-driven shifts in content demand.



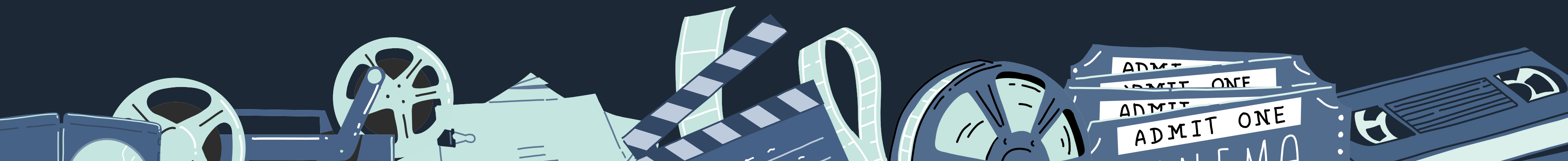
Objectives

- Integrate historical weather data with platform usage data
- Identify correlations between weather conditions and content engagement
- Use machine learning to predict high-demand and low-demand periods
- Simulate pricing strategies based on forecasted conditions
- Provide actionable recommendations for content scheduling and pricing



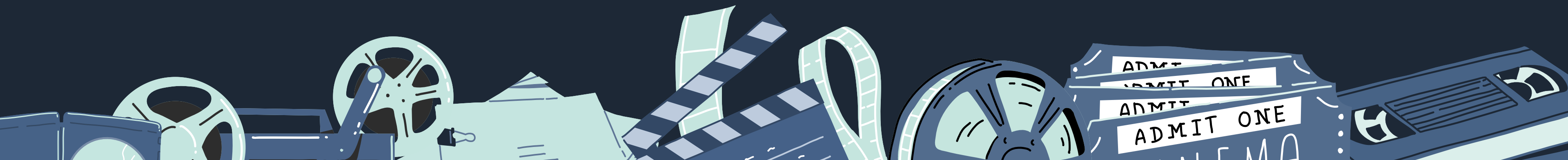
Data Understanding

- The data sourcing of the project included;
 - Viewing logs: Duration, timestamps, purchase type, content genre
 - Weather data: Historical temperature, rain, weather codes(via OpenMeteo API)
 - IP address to city mapping to get local climate context
- Time range – matches platform activity records to available weather data periods
- Merged datasets on location and date to enable correlation and prediction analysis



Feature engineering

- Data preparation – one hot encoding for categorical features, handling missing values
- Demand segmentation into High, Medium and Low based on view_minutes percentiles
- Used weather-based features like rainy_day flag(is_rainy), temperature bins(cold, mild, hot), precipitation and weather codes representing specific conditions



Modeling

- The demand level was predicted using a multi-class classification model, XGBoost and Prophet for viewership trends.
- Our target variable, “demand level” was categorized into three levels, “High”, “Medium” and “Low” using view time percentiles.
- The features for the model included the weather codes, temperature bins, precipitation etc
- We split the data 80/20 for training and testing. Evaluated the model through the accuracy score and feature importance to identify key demand drivers.



Dynamic Pricing Logic

- This is a rule-based systems that adjust content prices using the following conditions:
 - Demand-based adjustment: +20% for High, -10% for Low
 - Rainy day pricing and cold weather boost
 - Day of the week – more weights on weekends
- This strategy is designed to align prices with real-time demand drivers to maximize revenue and improve customer satisfaction.



Results

- Model performance:
 - We achieved model accuracy of 0.76 for the XGBoost classification model
 - The prophet model had MAE of 28.91 and RSME of 137.61
- The hybrid model improved demand forecasting by combining Prophet's long-term trend capabilities with XGBoost's capability to capture non-linear weather patterns.
- Integration of live weather data enables real-time simulation of future demand, making the system adaptive to changing external conditions

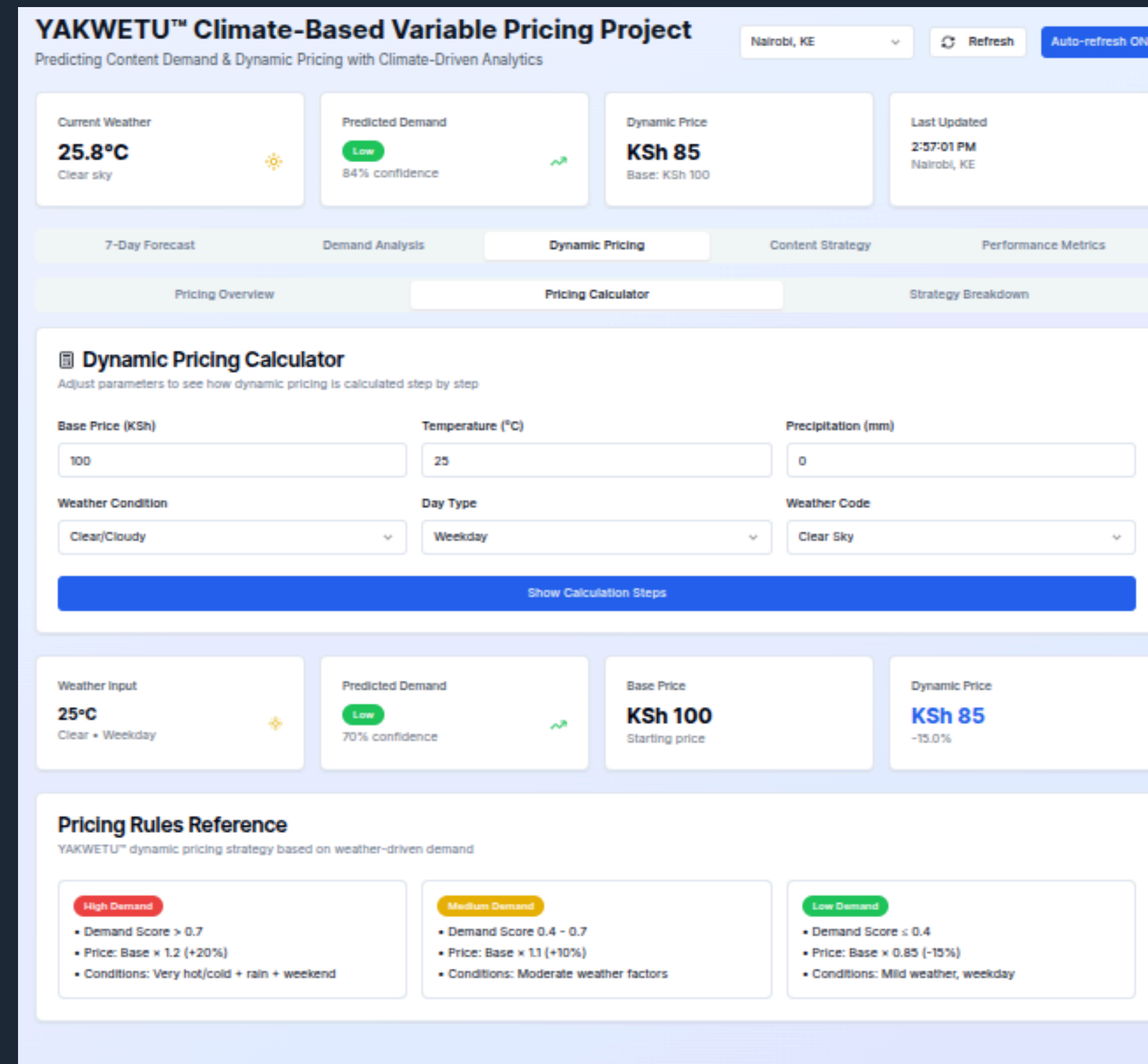
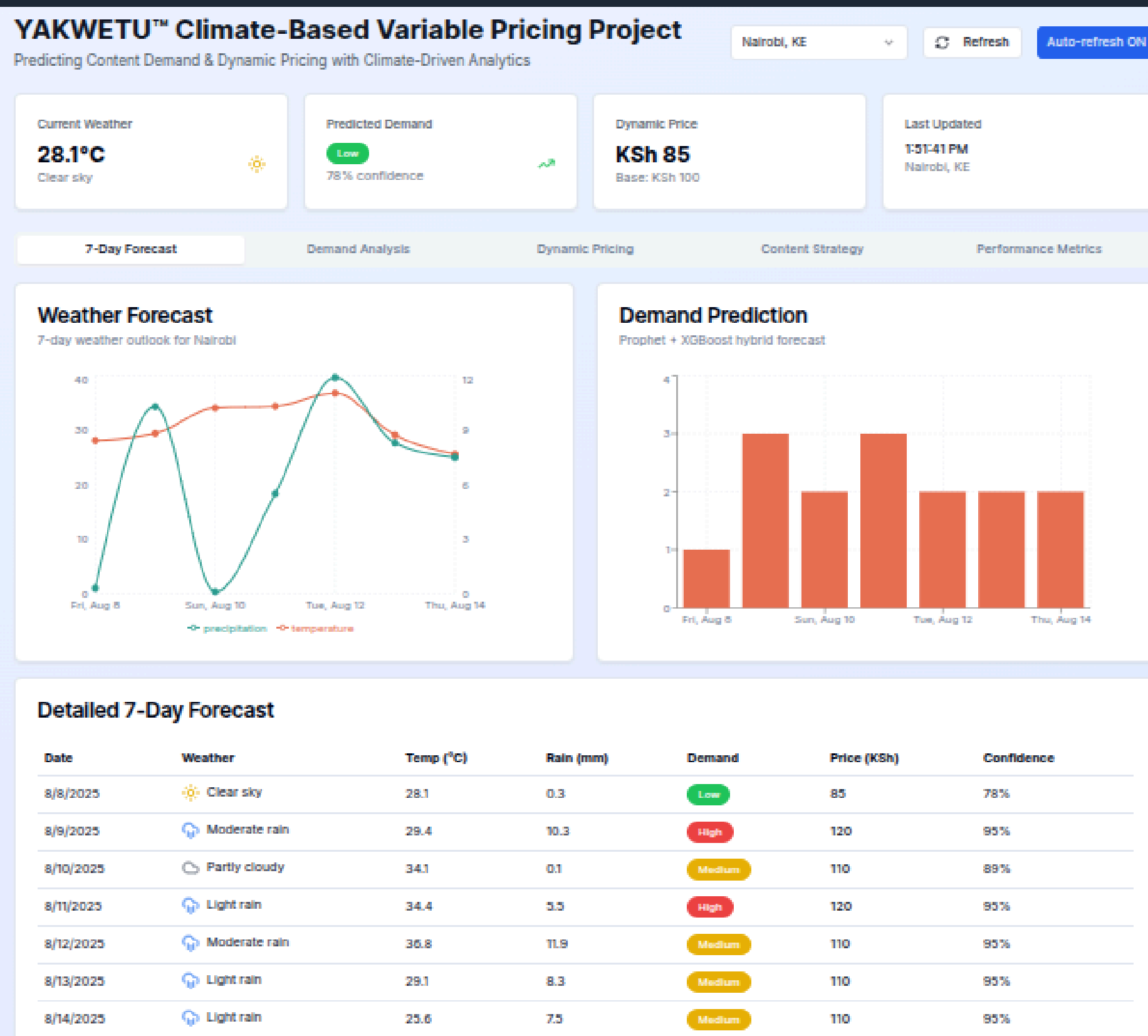


Results

Phase	Model	Accuracy	Key Innovation
<i>Phase 1</i>	XGBoost (initial)	~99.9% (leakage)	Identified data leakage risk
<i>Phase 1 (Fixed)</i>	XGBoost (leakage-free)	~45%	Clean, forecastable features only
<i>Phase 2 (Hybrid)</i>	Prophet + XGBoost	77.9% 	Combined time-series + ML intelligence



Deployment



Limitations

- Limited granularity of the weather as it was aggregated per city daily
- Assumptions in session reconstruction making us simulate view times
- The project definitely proves a correlation between weather and viewing patterns but other factors could be included in the model (e.g., holidays, special events, marketing campaigns) which could also be influencing demand.
- The prophet model when used alone captures general trends but it struggles with precise daily forecasts.



Recommendations

- Expand data collection on other potential influencing factors like special events or major holidays
- Granularity of viewing data – capturing the specific days/time people watch the content
- Retrain models regularly using the latest data or new techniques to maintain performance as trends shift.
- To mitigate risk, consider a phased rollout and gradually expand to other countries





Thank You

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